



PHANTOM 4.8 // DIAGRAM 08 — MULTI-CAMERA RING BUFFER & THREAD ISOLATION

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2
EXHIBIT

DECOUPLED 1-SLOT LATEST FRAME RING BUFFER ARCHITECTURE

Eliminates inference lag accumulation, guarantees zero buffer bloat, and provides complete camera thread isolation

MultiStreamYOLO26Manager ('backend/app/services/multi_stream_yolo26_manager.py')

CAMERA STREAM 01 // AHMEDABAD SG HIGHWAY (ACTIVE & HEALTHY)

CAPTURE THREAD #1

- RTSP over TCP Reader
- 25 FPS continuous pull
- Hardware PTS tracking

Status: **HEALTHY** (25 fps)

1-SLOT RING BUFFER #1

LATEST FRAME SLOT

Always holds frame N (Silent Overwrite)

- Queue length = EXACTLY 1
- ZERO BUFFER BLOAT (Sub-20ms)

INFERENCE WORKER THREAD

- Grabs latest frame from slot
- Executes YOLO26 + ANPR
- Inference latency: 16.4ms

Egress: **WebSocket Broadcast**

CAMERA STREAM 02 // SURAT RING ROAD (PACKET DROP / NETWORK TIMEOUT)

CAPTURE THREAD #2

- Network Socket Timeout
- 5.0s read deadline reached
- Backoff retry 2s -> 4s

Status: **RECONNECTING**

1-SLOT BUFFER ISOLATION

STALE SLOT DROPPED

Zero frames queued; no stall

- Thread sleeps peacefully
- ZERO IMPACT ON CAM 01 OR 03

SHARED GPU WORKER POOL

- Unaffected by Cam 02 drop
- Processes other 47 streams
- No deadlock, no GPU freeze

Worker health: **100% NOMINAL**

HARDWARE PTS DELTA DISCONTINUITY DETECTION & TRACK POOL RESET

THE VIDEO LOOP / PTS DRIFT PROBLEM

- Synthetic camera feeds or looped MP4/RTSP streams jump presentation timestamp (PTS) back to 0.0s
- In legacy systems, this causes severe identity drift:
Old vehicle track IDs fuse into newly looped cars
- Generates false trajectory alerts and corrupted dossiers

PHANTOM Delta Trigger:

```
if (current_pts - last_pts) < -1.0s:
```

AUTOMATIC TRACK POOL FLUSH

1. Timestamp Discontinuity Flagged:
`logger.warning("PTS jump detected on camera %s", cam_id)`
2. Clears Active Track Pool:
`tracker.reset_tracks(camera_id)`
3. Fresh Track IDs Allocated:
Next frame initiates clean tracking sequence

Result: **ZERO identity drift across looped feeds**

TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Decoupled 1-slot latest frame buffer guarantees zero buffer bloat and sub-25ms glass-to-memory capture latency.
- Hardware PTS delta tracking detects video loop restarts (pts_delta < -1.0s) and wipes track pool to avoid ID drift.
- Complete thread isolation prevents a dropped or lagging camera feed from stalling adjacent surveillance streams.